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Inventor(s)

Rivera; Jose Manuel et al.

System and Method for Secure Internet Communications

Abstract

Methods are disclosed for facilitating secure electronic communications that allow individual actors to be registered, their identities to be confirmed at an acceptable level of confidence, and their association with and/or ownership of certain user identifiers, to be verified. Secure and encrypted communication or digitally signed messages and/or documents between are supported while maintaining possession and control of one or more private cryptographic keys. To ensure that the integrity of information has not been compromised, embodiments are provided to locally generate and store private keys.

Inventors: Rivera; Jose Manuel (Burke, VA), Lasky; Michael (Silver Spring, MD), Bruce; Andrew (Glen Rock, PA)

Applicant: TripleCyber Corp. (Vienna, VA)

Family ID: 1000008628949

Assignee: TripleCyber Corp. (Vienna, VA)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/348,630, entitled “System and Method for Secure Internet Communications,” filed Jul. 7, 2023, which is a continuation of U.S. patent application Ser. No. 17/729,177, entitled “System and Method for Secure Internet Communications,” filed Apr. 26, 2022 (now U.S. Pat. No. 11,764,976, issued Sep. 19, 2023), which claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Patent Application No. 63/181,297, entitled “Method and System for Securing the Internet,” filed Apr. 29, 2021, the contents of which are incorporated herein by reference.

FIELD OF THE INVENTION

[0002] Embodiments of the present invention relate to an improved system and method of communicating on the Internet, where the identities of participants can be confirmed reliably, and where communications between participants can be secure. More particularly, embodiments of the present invention provide a new, more efficient, and more reliable computing and communications architecture for reliably registering Internet actors and confirming their identities, and for facilitating and managing secure Internet and network communications between those actors. Even more particularly, embodiments of the present invention provide a communications server and related software for registering and confirming the identities of individual actors on the Internet, and for facilitating communications between those individual actors at different levels of security and at different levels of identity confirmation. In addition, embodiments of the present invention also provide new user-device systems and methods that work together with a networked communications server to facilitate the creation, management, and use of public and private cryptographic keys to more efficiently enable secure Internet and network communications. Finally, embodiments of the present invention provide a secure method for recording user-level identifiers and their corresponding public keys on a publicly accessible, immutable cryptographic chain that can be used to verify the content and integrity of certain user-related information maintained on the communications server.

BACKGROUND

[0003] For all its eloquence and overall impact it has had on our lives, the current version of the Internet (circa 2021) has serious security flaws. Initially, these flaws were not viewed as flaws at all, because the Internet was intended to provide a mechanism for sharing information between disparate computing systems owned by entities that knew and cooperated with each other. It was conceived with the idea that collaboration was the primary goal, not security. Indeed, security was not a core feature of the Internet; it has always been an afterthought. Because of this, people have been trying to patch the Internet's endemically embedded security flaws, almost since the beginning.

[0004] One key security flaw, which enables bad actors to thrive on the Internet, is the ability to hide one's identity. It is almost trivial for anyone on the Internet to create a fake persona, including one that enables a user to masquerade as another person, a network, or even a fictitious entity.

While there may be situations where anonymity has advantages, there is no reason why anyone using a computing network, a cellular phone, or an email account should be required to accept communications from anonymous sources. Similarly, there is no reason why anyone should be required to accept communications from sources masquerading as others. And conversely, there are many times when someone might want to verifiably assert who they are.

[0005] With the proliferation of Internet domains, 4G-5G networks, and small devices capable of communicating on the Internet (i.e., the Internet of Things, or “IoT” devices), the importance of Internet security has risen dramatically.

[0006] In normal society outside the Internet, people have a semblance of protection based on what we want to share and how we want to share it. We can go into a restaurant, a store, or a theatre, and the information we share with the public can be selected and/or controlled, depending on the level of trust we have in the other party. We can share almost no information, paying with cash, for example, or we can share certain private information by paying with a credit card or other electronic means. We may be willing to share additional private information with a well-known and/or trusted party, while we may not be so willing to do so with someone who is not so well-known or trusted. Examples of trusted parties may include a bank, a Department of Motor Vehicles, or the Internal Revenue Service. An example of an untrusted party might include a stranger trying to sell Rolex watches on a street corner.

[0007] On the Internet, a cybercriminal may falsely appear as a trusted party. Once a cybercriminal penetrates a person's computing system, it is often far too late to do anything about it. Valuable digital assets can be copied, deleted, or rendered useless in milliseconds.

[0008] Historically, attempts to secure communications on the Internet have had difficulties. Many have involved a degree of complexity with respect to configuration and use that placed them out-of-reach for most users. Some attempts have required more active management and out-of-band transmission of passwords that degrade security and ease of use. Many attempts have required the storage of private keys and/or secure documents on a centralized server, where even though such documents are encrypted, there is nevertheless a suspicion that secrets might be exposed, either to the service provider or to a targeted attack. Given the importance of Internet security and the ease with which a malicious actor can masquerade as another identity, including a fake identity, and given the inadequacies in different solutions to this overall problem, there remains a need for an efficient system and method for enabling individual actors on the Internet to register themselves, to securely prove their identities to a sufficiently high level of probability, and to communicate with other Internet actors at a selected security level.

SUMMARY OF THE INVENTION

[0009] This summary is provided to introduce certain concepts in a simplified form that are further described below in the Detailed Description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to limit in any way the scope of the claimed invention.

[0010] Embodiments of the present invention overcome the shortcomings of prior art attempts to secure the Internet by solving several related problems.

[0011] In one aspect, a system is provided that includes an asymmetric key manager and one or more service modules to facilitate secure electronic communication. The asymmetric key manager is configured to generate a key pair including a first key pair having a first private key and a paired first public key. The first public key is assigned a security level configured to facilitate secured communication. The assigned security level is selected from one of public, private, secured, and enhanced. The asymmetric key manager is configured to store the generated first private key in a secure storage local to a user device. One or more service modules, operating local to the user device, are configured to utilize the asymmetric key manager to retrieve and use the first private key in support of one or more cryptographic functions and responsive to the assigned security level. The cryptographic functions include one or more of encryption, decryption, digital signature

generation, and digital signature verification.

[0012] Embodiments may assign each public key to a specific security level or purpose associated with a given user identifier, to enable the security of certain communications to be customized or tailored to specific applications and needs.

[0013] Embodiments may implement the communications server as a cloud-based server or a SAAS (software as a service) system, either of which may include software and digital ledger(s) and/or database(s) residing and/or executing on a plurality of Internet-accessible computing platforms.

[0014] In another aspect, a computer implemented method is provided to facilitate secure electronic communication. The embodiments include generating a key pair through the use of an asymmetric key manager. The generated key pair includes a first key pair having a first private key and a paired first public key. The embodiments further include assigning a security level to the first public, with the assigned security level configured to facilitate secured communication. The assigned security level is selected from one of public, private, secured, and enhanced. The generated private key is stored in a secure storage local to a user device. Embodiments further include retrieving and using the first private key in support of one or more cryptographic functions and responsive to the assigned security level. The cryptographic functions include one or more of encryption, decryption, digital signature generation, and digital signature verification.

[0015] Typically, embodiments will generate each public-private key pair for a specific purpose or security level. For example, an embodiment may generate a public-private key pair for the purpose of encrypting and decrypting email communications at a specified security level, or for signing and verifying signatures of email communications. This purpose or role, as well as other purposes or roles, will be discussed in greater detail below in the Detailed Description portion of this application.

[0016] When a transmitting, but unverified, actor on the Internet wishes to send a secure message to a verified actor whose user identifier and public key have been stored on the communications server, the transmitting actor may invoke embodiments to obtain the appropriate public key from the communications server (i.e., the public key associated with the target user identifier of the verified actor), use that public key to encrypt the message, and send the encrypted message directly to the target user identifier of the verified actor. When the verified actor receives the encrypted message, she may decrypt it using her corresponding private key.

[0017] Similarly, when a verified actor wishes to cryptographically sign a message that will be sent to a receiving actor (whether or not the identity of the receiving actor has been verified by the communications server), the transmitting verified actor need only sign the message with her private key, and then send the message to the receiving actor directly. When the receiving actor receives the signed message, the receiving actor may then interact with an embodiment of the communications server to obtain the appropriate public key associated with the transmitting verified actor's user identifier, and use the obtained public key to verify that the signature was created by the transmitting verified actor. The receiving actor may also obtain information about the transmitting verified actor from the communications server, in order to confirm that the transmitting verified actor is who he says he is.

[0018] Third, to ensure that the integrity of the information on the communications server has not been compromised, embodiments of the communications server may periodically create a backup of crucial public data—for example, user identifiers and their associated public keys—and save such backups in a format that cannot reasonably be altered. For example, an embodiment may create a backup comprising a cryptographic chain of a plurality of chained records obtained from the digital ledger, where each chained record includes a hash or hashes of the user identifier of a verified actor and the public key associated with that user identifier. An embodiment may then escrow a copy of this cryptographic chain to a publicly available, immutable, append-only Internet blob, the contents of which can be cross-referenced with and validated against corresponding data

residing in the digital ledger on the communications server.

[0019] Using the above-summarized solutions, embodiments may ensure that digital messages can be sent and received securely via the Internet and other networks to and from actors and devices whose identities have been verified, thereby significantly reducing the probability of malicious actions. Fictitious and/or malicious actors may still exist on the Internet; but their existence will be obvious because their identity will not have been verified, for example. Embodiments may thereby ensure that if an Internet actor wishes to communicate through an unverified persona, it can, while simultaneously ensuring that when the actor wants to communicate in a secure manner as a verified identity, it can do that as well.

[0020] The above summaries of embodiments have been provided to introduce certain concepts that are further described below in the Detailed Description. The summarized embodiments are not necessarily representative of the claimed subject matter, nor do they span the scope of features described in more detail below. They simply serve as an introduction to the subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] So that the above recited features of the present invention can be understood in detail, a more particular description of the invention may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of the invention and are therefore not to be considered limiting of its scope, for the invention may admit to other equally effective embodiments.

[0022] FIG. 1 is a block diagram illustrating an exemplary embodiment of a communications server for enabling secure Internet communications, in accordance with the present invention.

[0023] FIG. 2 is a block diagram illustrating an exemplary embodiment of a Database Management System (DBMS) component of a communications server for enabling secure Internet communications, in accordance with the present invention.

[0024] FIG. 3 is a combined flow chart and block diagram illustrating an exemplary embodiment of a method for registering users to enable secure Internet communications, in accordance with the present invention.

[0025] FIG. 4 is a combined flow chart and block diagram illustrating an exemplary embodiment of a method for providing public key services to enable secure Internet communications, in accordance with the present invention.

[0026] FIG. 5 is a combined flow chart and block diagram illustrating an exemplary embodiment of a public-private key manager (referred to herein as a “Trifold” and defined in greater detail below) for enabling secure Internet communications, in accordance with the present invention.

[0027] FIG. 6 is a combined flow chart and block diagram illustrating an exemplary embodiment of a method for verifying a user's identity to enable secure Internet communications using a prepared URL, in accordance with the present invention.

[0028] FIG. 7 is a combined flow chart and block diagram illustrating an exemplary embodiment of a method for verifying a user's identity to enable secure Internet communications using an SMS message delivered to a user's cellphone, in accordance with the present invention.

[0029] FIG. 8 is a combined flow chart and block diagram illustrating an exemplary embodiment of a method for verifying a user's identity to enable secure Internet communications using a trusted name-address source, in accordance with the present invention.

[0030] FIG. 9 is a combined flow chart and block diagram illustrating an exemplary embodiment of a method for encrypting user data, in accordance with the present invention.

[0031] FIG. 10 is a combined flow chart and block diagram illustrating an exemplary embodiment

of a method for decrypting user data, in accordance with the present invention.

[0032] FIG. **11** is a combined flow chart and block diagram illustrating an exemplary embodiment of a method for cryptographically signing a digital document, in accordance with the present invention.

[0033] FIG. **12** is a combined flow chart and block diagram illustrating an exemplary embodiment of a method for verifying a digital signature, in accordance with the present invention.

[0034] FIG. **13** is a block diagram of an exemplary embodiment of a computing device, in accordance with the present invention.

DESCRIPTION OF THE EMBODIMENTS

[0035] Embodiments of the present invention will be described with reference to the accompanying drawings, wherein like parts are designated by like reference numerals throughout, and wherein the leftmost digit of each reference number refers to the drawing number of the figure in which the referenced part first appears. Embodiments of the invention are not limited in their application to the details of construction or to the arrangement of the components set forth in the following description or illustrated in the drawings. Embodiments of the invention are capable of being practiced and carried out in various ways. In addition, the phraseology and terminology employed herein are for the purpose of description and should not be regarded as limiting.

Communications Server

[0036] FIG. **1** is a block diagram illustrating an exemplary embodiment of a Communications Server **100** for enabling secure Internet communications, in accordance with the present invention.

Communications Server **100** is a logical construct that may be implemented in a number of physical configurations known to one skilled in the art, but each configuration will comprise the equivalent of an Internet-accessible computing server, similar to a web server with a database management system. Communications Server **100** may provide certain capabilities and functions that together enable users to communicate with a high degree of confidence concerning the identity of other users. To provide its capabilities and functions, embodiments of the Communications Server **100** may include a Database Management System **105**, a Trifold Registration Service **120**, a User Registration Services Module **130**, a Public Key Services Module **140**, and a Cryptographic Chain Services Module **150**.

[0037] The term “Trifold,” as used herein, refers to a software module installed on a user's device. A Trifold, such as Trifold **173** and/or Trifold **183**, may perform public-private key creation and management functions, as well as cryptographic functions (such as encrypting a message or signing a message), on behalf of requesting applications. A Trifold may communicate with the Communications Server **100**, may create public-private key pairs, may provide access to previously created public and private keys, and may perform other related tasks that will be discussed in more detail below. Trifold Registration Service **120** may interact with Trifolds (e.g., Trifold **173** and/or Trifold **183**) residing on user devices.

[0038] Continuing to refer to FIG. **1**, Database Management System **105** may include tables of User Related Records **110**, where each such user-related record may contain information relating to a single user, and may also store (a) public keys for use in cryptographic communications with a registered device, (b) information about computing devices registered to a user, and (c) information about applications that may reside and execute on a user's computing device, which may be registered to a user. An application may be referred to herein as an “asset.”

[0039] Database Management System **105** may interact with the User Registration Services Module **130** to store and retrieve information about registered users. The User Registration Services Module **130** may perform these functions by providing a set of in-code data entities (e.g., structures, objects, etc.) for user data that are conformable with corresponding data stored in the Database Management System **105**. The User Registration Services Module **130** may use industry standard database libraries (e.g., Microsoft's entity framework) to provide a connection between these in-code entities and the data stored in Database Management System **105**. Data from

Database Management System **105** may be fetched into the data entities maintained by User Registration Services Module **130**, manipulated there, and then saved back to the Database Management System **105** using the provided capabilities of the library. For example, the User Registration Services Module **130** may use Microsoft's Entity Framework and LINQ libraries to move data between SQL Server database tables and C #objects that conform to these tables.

[0040] Database Management System **105** may interact with the Public Key Services Module **140** to provide public keys to other users (for example, to Requester **185**) upon request. The Public Key Services Module **140** may perform these functions by providing a set of in-code data entities (e.g., structures, objects, etc.,) for key data that are conformable with corresponding data stored in Database Management System **105**. Public Key Services Module **140** may use industry standard database libraries (e.g., Microsoft's entity framework) to provide a connection between these in-code entities and the data stored in Database Management System **105**. Data from Database Management System **105** may be fetched into the data entities maintained by Public Key Services Module **140**, manipulated there, and then saved back to the Database Management System **105** using the provided capabilities of the library. For example, Public Key Services Module **140** may use Microsoft's Entity Framework and LINQ libraries to move data between SQL Server database tables and C #objects that conform to these tables. Public Key Services Module **140** may provide a well-defined API through which an external software process (i.e., another program, such as a Trifold **173**) can submit key data for storage to Database Management System **105**, or another program (such as a Trifold **183**) can submit a request (such as Request **187**) for a particular key corresponding to a particular user registered in Database Management System **105**. (A response to that request, for example Response **189**, may provide the requested key.) For example, Public Key Services Module **140** might implement a restful web service interface that accepts query parameters in a Get query and returns data in JSON format.

[0041] Database Management System **105** may interact with the Cryptographic Chain Services Module **150** to create Cryptographic Chain Data **115** and to store a backup of Cryptographic Chain Data **115** to a remote, Append-Only, Immutable Blob **190** via the Internet **160**. Cryptographic Chain Services Module **150** may include in the backup of Cryptographic Chain Data **115** a backup of certain data stored in User Related Records **110**.

[0042] The Cryptographic Chain Services Module **150** may perform these functions by providing a set of in-code data entities (e.g., structures, objects, etc.,) that represent the cryptographic chain of hashes commonly used to create an immutable record. Such chains are generally referred to as block chains and are well understood by those skilled in the art. The data structures for these chains are conformable to the data stored in Database Management System **105**. The Cryptographic Chain Services Module **150** may use industry standard database libraries (e.g., Microsoft's entity framework) to provide a connection between these in-code entities and the data stored in Database Management System **105**. Data from Database Management System **105** may be fetched into the data entities, manipulated there, and then written back to the Database Management System **105** using the provided capabilities of the library. For example, the Cryptographic Chain Services Module **150** may use Microsoft's Entity Framework and LINQ libraries to move data between SQL Server database tables and C #objects that conform to those tables. The Cryptographic Chain Services Module **150** may use industry standard cryptographic libraries to compute the desired hashes of the Public Key and owning User Identifier data that are tracked and to add them to then end of Cryptographic Chain Data **115** for saving in the Database Management System **105**. For example, the system may use cryptographic libraries that are part of Microsoft's .NetCore Framework.

[0043] The Cryptographic Chain Services Module **150** may escrow Cryptographic Chain Data **115**, certain hashed data retrieved from User Related Records **110**, and the constructed hash chains to a remote data storage location that is established as an Append Only Immutable Blob **190**, which is not subject to subsequent manipulation by the Communication Server **100** or any software therein.

That is, Append Only Immutable Blob **190** is a storage medium maintained by a third party that provides a medium that permits writing, but does not allow updating. For example Append Only Immutable Blob **190** may correspond to a Blob offered on the Azure cloud. The Cryptographic Chain Data **115** and certain hashed data retrieved from User Related Records **110** may be written to Append Only Immutable Blob **190** using the functions within an API provided by the third-party provider of Append Only Immutable Blob **190** storage. For example the system may use the Azure.Storage.Blobs library provided by Microsoft to write data to Azure Immutable Blobs.

[0044] Database Management System **105** may interact with the Trifold Registration Service **120** to store (a) information related to a user's Trifold (such as Trifold **173** or Trifold **183**), (b) information about the user's devices where a Trifold resides (such as User Device **170** or Requester Device **180**), and/or (c) information about various software applications running on a user's device (such as User Device **170** or Requester Device **180**) that may be interacting with a Trifold. The Trifold Registration Service **120** may perform these functions by providing a set of in-code data entities (e.g., structures, objects, etc.,) corresponding to devices, applications, and Trifold client certificates that are conformable with corresponding data stored in Database Management System **105**. The Trifold Registration Service **120** may use industry standard database libraries (e.g., Microsoft's Entity Framework) to provide a connection between these in-code entities and the data stored in Database Management System **105**. Data (received from a remote Trifold such as Trifold **173** or Trifold **183**) in these data entities may be written to the Database Management System **105** using the provided capabilities of a respective library. The Trifold Registration Service **120** may provide a well-defined API through which a remote Trifold (such as Trifold **173** or Trifold **183**) can provide data. For example the Trifold Registration Service **120** may implement a restful web service interface that accepts submitted data as query parameters. A registering Trifold (e.g., Trifold **173**) may provide to the Trifold Registration Service **120** the type of User Device **170** and the name of User Device **170**, along with a serial number. A registering Trifold (e.g., Trifold **173**) may also provide to the Trifold Registration Service **120** an application identifier that was issued to a corresponding application developer when a given application type was registered for interacting with a Trifold. In response, a Trifold (such as Trifold **173**) may receive a TLS (transport layer security) certificate (generated using standard cryptographic libraries) that is tied explicitly to a specific Trifold for a specific registered user (e.g., User **175**), for a specific device (e.g., User Device **170**), and for a specific application (not shown), to secure all future interactions with the Communication Server **100**.

[0045] User Registration Services Module **130** is a software module executing on the Communications Server **100** that may perform the following functions: (1) register new users (such as User **175**); (2) verify the identity of users using a variety of methods described below; and (3) determine whether a user's identity is known with sufficient confidence to allow that user to communicate at one of several available security levels. User Registration Services Module **130** may communicate with registering users such as User **175** via User Device **170** via the Internet **160**. The information received from the User **175** is related to the verification of their digital identifiers with their real-world identities. For example, a user's digital identifier may be an email address or a phone number. Real-world identity information may comprise a given and family name of the User **175** and may also include their physical (street, city, etc.) address. Depending on the desired confidence level for identity verification and whether the provided identity information can be programmatically matched with trusted outside sources (discussed in more detail later) a user's identity can be verified. At higher levels of security, additional name and address information may be provided by a user in the form of selection of names and addresses from automatically generated lists, electronic scans of trusted government documents (e.g., a driver's license) or Geo-Location coordinates to test for presence at a provided location. Additionally, a real-time video of a user may be provided to allow for image liveness testing.

[0046] Data is also obtained by the User Registration Service **130** from various external "Trusted"

sources to assist in the validation of the provided user name and address. Commonly used industry providers of Address verification and phone ownership may be queried using an API provided by these vendors to assess the veracity of the provided name and address. CNAM databases from phone providers may be checked as well.

[0047] Public Key Services Module **140** is a software module executing on the Communications Server **100** that may perform the following functions: (1) receive requests for a public key for a specific user for a specific purpose (for example, receive a request from Requester **185** for a public key associated with User **175** for encrypting a message); (2) extract a public key from the Database Management System **105** for a given user (such as for User **175**); and (3) transmit an extracted public key corresponding to a given user (such as User **175**) to a requesting user (such as Requester **185**). Public Key Services Module **140** provides a well-defined API through which external software process (i.e., another program such as Trifold **173**) can request key data for one or more users for one or more uses. For example, Public Key Services Module **140** might implement a restful web service interface that accepts query parameters for a particular user and a particular key type as query parameters of a Get query and returns the requested public key as a Base64 encoded string in a standard JSON payload.

[0048] Cryptographic Chain Services Module **150** is a software module executing on the Communications Server **100** that may perform the following functions: (1) interact with the Database Management System **105** to assemble cryptographically linked chains of selected hashed Cryptographic Chain Data **115**; and (2) escrow a copy of the cryptographically linked chains to a remote Append-Only Immutable Blob **190** on the Internet **160**. Third parties will be able to validate the long term ownership of a public key to a particular user against the cryptographic chains in two ways, depending on their degree of interest. Most easily they can compare the value of a key they have received against the value in the cryptographic chains that are stored in Database Management System **105** and can be requested from Cryptographic Chain Services **150** API. The Cryptographic Chain Services **150** API may implement a web service that provides both the Cryptographic Chain Data **115** and the data from which it was generated (obtained from Database Management System **105**) as a CSV file (for example). The exact specification of how hashes are constructed and chained are publicly available and well known by those skilled in the art. For example the Communication Server **100** may provide a web page where these details are displayed. This allows validation of the integrity of the Cryptographic Chain Data **115** and allows verification of the value and ownership of any key therein. With slightly more difficulty, a user could choose to perform the same validation by downloading the escrowed data from the third party provided immutable blobs (for example, the escrowed data in Append Only Immutable Blob **190**). Third party providers are selected in part, on their ability to make this action easily available. For example, Azure immutable blobs may be used, which directly provide a CSV download in response to an HTTPS request to a specific endpoint. Finally a third party can compare the data available from Database Management System **105** against the data available from the escrowed storage (Append Only Immutable Blob **190**) and check that they are identical. For example, if the both the download from Database Management System **105** and escrowed data from Append Only Immutable Blob **190** are provided in identical CSV format, then an identity comparison can easily be performed by any user using a spreadsheet.

[0049] A User **175** may interact with Communications Server **100** via a User Device **170** to communicate with the Communications Server **100** via the Internet **160**. User **175** may first interact with Communications Server **100** via a User Device **170** to become a registered user. User Device **170** may be a typical computing device capable of interacting with Internet web pages and the like. User Device **170** may also be a smartphone or similar device. User device may have Trifold **173** installed during user registration, or the Trifold **173** may be obtained later.

[0050] Becoming a registered user is a multi-step process that will be described in more detail below. Once User **175** is registered, however, the Communications Server **100** will have verified

the identity of User **175** at one of several different confidence levels (or security levels) using methods described in more detail below. Once the User **175** is registered at some security level, a Trifold **173** can be installed and registered for User **175** on User Device **170**. Such registration may result in the Communications Server **100** issuing a TLS (transport layer security) client certificate for the Trifold **173** to use later for securing future communications. A TLS certificate is an industry standard format for packaging a private key along with other identity information (a subject name) in a password secured package. The public key of the TLS certificate may be maintained on the Communication Server **100** (and is tied to a particular Trifold on a device for an application, for a user). The TLS certificate itself (with an encrypted private key) may be transferred to the Trifold (such as Trifold **173**) where it can be stored. The password for the TLS certificate (a secret) may be stored in a secure fashion for a user (such as User **175**) on the user's device (such as User Device **170**) by the Trifold (such as Trifold **173**). Such registration may also authorize a Trifold **173** to generate public and private key pairs to facilitate communication between User **175** and other registered users, for example Requester **185**.

[0051] If another user, such as Requester **185**, wishes to send a message to User **175**, Requester **185** may cause Trifold **183** on Requester Device **180** to send a secured (using a TLS client certificate) Request **187** via the Internet **160** to the Public Key Services Module **140** of Communications Server **100**. Request **187** may be sent using any number of standard protocols known to those skilled in the art. For example, Request **187** may be sent using a restful HTTPS query. Request **187** may ask Communications Server **100** to provide a public key for sending encrypted messages to User **175**. If User **175** and its Trifold **173** have been registered on Communications Server **100** such that an appropriate public key has been generated and stored in an appropriate User Record **110** of Database Management System **105** for User **175**, then in response to Request **187**, the Public Key Services Module **140** of Communications Server **100** may extract the requested public key from the appropriate User Record **110** and transmit it to Requester **185** via the Internet **160** to Trifold **183** on Requester Device **180**. Using the public key provided in the Response **189** to Request **187**, the Requester **185**'s Trifold **183** may send an Encrypted Message **165** to User **175**. When the Encrypted Message **165** is received by User **175**, User **175** may use Trifold **173** to decrypt Encrypted Message **165** using User **175**'s private key corresponding to the public key that was used to encrypt the Encrypted Message **165** sent by Requester **185**.

Database Management System

[0052] FIG. **2** is a block diagram illustrating an exemplary embodiment of a Database Management System (DBMS) **200** component of a Communications Server **100** for enabling secure Internet communications, in accordance with the present invention. DBMS **200** may be an instantiation of Database Management System **105** shown in FIG. **1**.

[0053] Referring to FIG. **2**, DBMS **200** may be modeled using any number of data models (e.g., relational, network, etc.) common to those skilled in the art. Similarly, any number of commonly available database management systems (e.g., SQL server, Oracle, Raima Database Manager) familiar to all practitioners could be used to instantiate the selected model. For consistent case of presentation, without restricting the scope of the invention, DBMS **200** will be described using terminology and concepts (tables, records, fields, keys) based on the relational data model.

[0054] Within the DBMS **200** are User Related Records **210**, which are records containing data that relate either directly or indirectly to individual users. User Related Records **210** may be an instantiation of User Related Records **110** shown in FIG. **1**. As shown in FIG. **2**, the data stored in User Related Records **210** may comprise the following tables and/or records: User Verification Info **221**, User Identifier **222**, Encryption Public Key **223**, Signature Public Key **224**, Client Certificate **225**, Security Level **226**, and User Assets **227**.

[0055] User Verification Info **221** may contain information describing operational results of an identity verification process within the User Registration Services **300** shown in FIG. **3** and described in detail below. Still referring to FIG. **2**, however, User Verification Info **221** may

provide a record of the basis for which verification at a particular security level was granted. It may also provide information on the exact location of a user within a non-completed identity verification process.

[0056] User Identifier **222** may contain data used to identify a user. User Identifier **222** may include a globally unique user-name. It may also contain additional identifying user information such as an email address, a phone number, or a given name or family name.

[0057] Encryption Public Key **223** may contain data related to the public key of a remotely generated public/private key pair tied to a particular user. Encryption Public Key **223** may be stored in a format appropriate to the selected DBMS **200**. It may contain additional information such as a thumbprint, a validity date range, and additional meta data. Encryption Public Key **223** should be of sufficient strength to be suitable for encryption of messages sent via the Internet. As a cryptographic best practice familiar to all those skilled in the art, Encryption Public Key **223** will preferentially be distinct from Signature Public Key **224**.

[0058] Signature Public Key **224** may contain data related to the public key of a remotely generated public/private key pair tied to a particular user. Signature Public Key **224** may be stored in a format appropriate to the selected DBMS **200**. It may contain additional information such as a thumbprint, a validity date range, and additional meta data. The encryption key strength of Signature Public Key **224** may be less than that of Encryption Public Key **223**. As a cryptographic best practice familiar to all those skilled in the art, Signature Public Key **224** will preferentially be distinct from Encryption Public Key **223**.

[0059] Client Certificate **225** may contain information about a public key for the TLS (Transport Layer Security) client certificate that was issued by Communications Server **100** to allow trusted communication to occur between Communications Server **100** and a user's Trifold, such as Trifold **173**. Depending on exact implementation, Client Certificate **225** may contain additional data identifying the particular user device (e.g., User Device **170**) and particular trifold (e.g., Trifold **173**) to which it was issued. Client Certificate **225** may also contain expiration information allowing for easy invalidation in the event a user device (such as User Device **170**) is compromised.

[0060] Security Level **226** may contain information about a derived or granted security level of the user (such as User **170**) based on the findings of the user verification process described below. For example, Security Level **226** may correspond to a security level of “public,” “private,” “secure,” or “enhanced.” Security Level **226** may be referenced by software running on Communications Server **100** (for example, in Public Key Services Module **140**) to grant or restrict certain requests made by User **175** or Requester **185**, via software executing on User Device **170** or Requester Device **180**, respectively.

[0061] User Assets **227** may contain information related to the assets registered to a user (such as User **170**), or assets that interact with Communication Server **100** or are tracked for security verification. In this context, the term “assets” has a generic meaning that may include computing devices, software applications, Trifolds, Internet domains, and the like.

User Registration Service

[0062] FIG. **3** is a combined flow chart and block diagram illustrating an exemplary embodiment of a method for registering users to enable secure Internet communications, in accordance with the present invention.

[0063] Referring to FIG. **3**, User Registration Service **300** comprises a general process through which a user may be registered on Communications Server **100**. User Registration Service **300** may be an instantiation of the User Registration Service **130** Module discussed with respect to FIG. **1**. User Registration Service **300** may include the following process steps: Receive and Store User Identifier **310**, Verify User Identity **320**, and Set and Store User Security Level **330**.

[0064] The step corresponding to Receive and Store User Identifier **310** may comprise a process in which user identifiers (email addresses and the like) are received via a user registration interaction

identified as Registration **340**, and are stored within the Database Management System **105** (also DBMS **200**). Registration **340** process may occur through a simple web page interface. It may involve a single user such as User **305** providing a single identifier such as an email address. It may also involve a user such as User **305** providing multiple identifiers for multiple users. This step may also establish connection credentials for User **305**. User **305** may be an instantiation of User **175** shown in FIG. **1**.

[0065] The step corresponding to Verify User Identity **320** may comprise a process where a user's identity is verified. A confidence level (i.e., the probability of accuracy) of an identity verification may be tied to a corresponding security level. That is, the higher the confidence of accuracy in a user's identity, the higher the security level that can be granted. The various stages of user identity verification are explained in more detail below. The Verify User Identity **320** process may involve direct interaction with User **305**. It may involve User **305** using a device such as User Device **370** to respond to URL links embedded in email. It may involve User **305** relaying codes transmitted by SMS. It may involve User **305** selecting from possible options or responding to questions presented on a web page.

[0066] The step corresponding to Set and Store User Security Level **330** may comprise a process where the results of the Verify User Identity **320** process are evaluated and stored as a security level for the user (such as User **305**). This security level may result in User **305** being granted or denied certain activities or requests later when the User **305** uses or attempts to use any of various software applications residing and/or executing on User Device **370**. For example, the security level may be referenced and used to grant or deny User **305** the ability to purchase certain products.

Public Key Services

[0067] FIG. **4** is a combined flow chart and block diagram illustrating an exemplary embodiment of a method for providing public key services to enable secure Internet communications, in accordance with the present invention.

[0068] Referring to FIG. **4**, Public Key Service **400** illustrates the overall process where public keys are provided to users who request them. Such a Request **460** may originate from a remote user device such as User Device **470** controlled by a user such as User-Requester **405**. Request **460** may originate from a Trifold **475** executing on User Device **470**. Request **460** may originate from some other user-controlled device as well. Also, Trifold **475** may issue Request **460** because it received an upstream request for public keys from an application (not shown) running on User Device **470**.

[0069] The step corresponding to Receive Public Key Request **410** comprises a listening step in which a Request **460** to provide a public key is received from User Device **470** on behalf of User-Requester **405**. Request **460** may request a public key corresponding to a given user and may also be for a particular purpose, for example, to provide the signature verification public key (such as Signature Public Key **224** shown in FIG. **2**) for the given user. Request **460** may also indicate a particular purpose for a set of users, for example to provide an encryption key (such as Encryption Public Key **223** shown in FIG. **2**)) for User A and User B.

[0070] The step corresponding to Extract Requested Public Key Request **420** comprises a process where the defining qualifications that were received in Receive Public Key Request **410** are transformed into query parameters and used to extract the requisite public keys from the DBMS **200** (discussed above in the context of FIG. **2**). These parameters may include single or multiple user identifiers and single or multiple purposes.

[0071] The step corresponding to Transmit Requested Public Key **430** comprises a process where the Public Key **440** (which may comprise several such public keys) extracted in the previous Extract Requested Public Key Request **420** step is returned to the requesting Trifold **475**. Keys such as Public Key **440** are returned in a format consistent with a supported calling protocol known to those skilled in the art.

User Public Private Key Manager

[0072] FIG. **5** is a combined flow chart and block diagram illustrating an exemplary embodiment

of a Trifold **530** (a user-public-private key manager) for enabling secure Internet communications, in accordance with the present invention.

[0073] FIG. 5 also illustrates the Communication Server **550** (discussed previously as Communication Server **100** in FIG. 1), User Device **510** (discussed previously as User Device **170** in FIG. 1), and the Trifold Registration Service **555** (discussed previously as Trifold Registration Service **120**).

[0074] Referring to FIG. 5, Trifold Registration Service **555** may be responsible for the initial registration and authorization of a Trifold **530** executing on a physically remote User Device **510**. Trifold Registration Service **555** may receive a Trifold Identifier **590** that identifies a particular Trifold **530** for a particular User Device **510**. The Trifold registration service may accept an industry standard JWT authentication token from an industry standard OAuth server for authentication of this user. This process is familiar to practitioners of the art. Trifold Registration Service **555** may then store Trifold Identifier **590** in the appropriate User Related Records of Database Management System **560** (corresponding to the User Related Records **210** of DBMS **200** in FIG. 2 discussed above). Using common cryptographic libraries familiar to all practitioners, Trifold Registration Service **555** may generate a TLS Client Certificate **570** that will be used by Trifold **530** to identify itself in all future communications with the Communications Server **550**. Client Certificate **570** is a commonly used package for password protected private keys. The Certificate itself, which is returned to the Trifold **530**, can simply be stored where convenient on the file system of Communication Server **550**. The Client Certificate Password **571** for Client Certificate **570** is a secret, and the Trifold **530** should preferably store the corresponding Client Certificate Password **571** within the Secured Storage **540**. The public key associated with the private key of the Client Certificate **570** is communicated to the Database Management System **560**. It is stored in its own table, relationally tied to the relevant user (not shown), the User Device **510**, and the User Application **520** for which the Trifold **530** was registered. The Trifold **530** on User Device **510** may then generate the public/private key pairs that are required. A Public Key **580** may be transmitted to the Communication Server **550** using the Public Key Services **400** (discussed earlier and shown in FIG. 4) for storage in the Database Management System **560**. The Private Key **581** of the pair are stored in the Secure Storage **540** on User Device **510**.

[0075] User Device **510** corresponds to User Device **170**, User Device **370**, and User Device **470**. It may include User Application **520**, Trifold **530**, and the Secure Storage **540**.

[0076] User Application **520** may comprise a software application configured to communicate with Trifold **530**. Being “configured to communicate with Trifold **530**” means User Application **520** may be programmed to invoke the following services of Trifold **530**: Encrypt **533**, Decrypt **535**, Sign **537**, and/or Verify **539**. It may also have the capability to interact with an industry standard OAuth server and transfer a returned industry standard JWT authentication token to the Trifold.

[0077] Trifold **530** comprises a component software module operating on User Device **510** that performs various public-private key management functions, including (a) public-private key generation; (b) public-private key management; and (c) public-private key use. These functions are organized as follows: Initialize **531**, Encrypt **533**, Decrypt **535**, Sign **537**, and Verify **539**.

[0078] The Initialize **531** function comprises initial preparation and configuration functions that should be performed before the Trifold **530** can perform other functions. Initialize **531** automatically executes the first time Trifold **530** is invoked. Initialize **531** calls, via the Internet, the Trifold Registration Service **555** discussed above, which may reside and execute on the Communication Server **550**. In response, Trifold Registration Service **555** may provide a Trifold Identifier **590** corresponding to User Device **510** and Client Certificate **570** corresponding to Trifold **530**. Initialize **531** receives the Client Certificate **590**, which it may use to identify itself for all future communications with the Communications Server **550**. Initialize **531** then generates the public-private key pairs for signature and encryption purposes.

[0079] Initialize **531** may then transmit Public Key **580** portions of the generated public-private

key pairs to the TrifoldRegistration Service **555** executing on the Communication Server (**550**) for storage in the Database Management System **560** as discussed above.

[0080] For each Public Key **580** generated by the public-private key pair generation process executing within the Initialize **531** function, a corresponding Private Key **581** is generated. Both Private Key **581** and Client Certificate's Password **571** may be securely stored in Secure Storage **540**. The exact location and mechanism for securely storing data are different on each supported operating system, but those locations and mechanisms are familiar to those skilled in the art. As the Client Certificate's private key is encrypted, it can simply be stored on the device's file system in a convenient location.

[0081] Encrypt **533** is a process that implements the encryption capabilities of the Trifold **530**. It is described in detail below with respect to FIG. **9**.

[0082] Decrypt **535** is a process that implements the decryption capability of the Trifold **530**. It is described in detail below with respect to FIG. **10**.

[0083] Sign **537** is a process that implements the digital signature generation capability of the Trifold **530**. It is described in detail below with respect to FIG. **11**.

[0084] Verify **539** is a process that implements the digital signature verification capability of the Trifold **530**. It is described in detail below with respect to FIG. **12**.

User Identity Verification Using URLs

[0085] FIG. **6** is a combined flow chart and block diagram illustrating an exemplary embodiment of a method for verifying a user's identity to enable secure Internet communications using a prepared URL, in accordance with the present invention.

[0086] Referring to FIG. **6**, the User Identify Verification Method **600** depicted in FIG. **6** illustrates how a user's email address may be verified by embodiments of the present invention. In Step **610**, an email message containing a prepared URL may be sent via an email message to an email address provided by User **605**. The email message may instruct User **605** to click on the embedded URL, which, when clicked or selected, will preferably direct User **605** to a prepared web page hosted on a communications server such as Communications Server **100** (see FIG. **1**). The URL may additionally contain a prepared token specific to this particular User **605** for this particular purpose at this particular time.

[0087] Using User Device **620**, User **605** may open the email sent in Step **610** using any industry standard email client. The User **605** may respond by clicking on the URL embedded therein. The clicking action may invoke a web browser displayed on User Device **620** and cause the web browser to open the web page indicated by the URL. When the web page opens, code on the web page may notify software executing on the Communications Server **100** (see FIG. **1**) that the Prepared URL was accessed within a specified period of time. At Step **630**, upon receiving notice that the user activated the required URL within the specified response time, Step **630** may then invoke Step **640** to record the fact that User **605** has been verified to use the provided email address. Such verification may be manifested by a message transmitted to the user and/or by recording an indication of user identity verification at a corresponding security level within the Database Management System **105** described above with respect to FIG. **1** and/or DBMS **200** described above with respect to FIG. **2**.

User Identity Verification Using SMS Messages

[0088] FIG. **7** is a combined flow chart and block diagram illustrating an exemplary embodiment of a method for verifying a user's identity to enable secure Internet communications using an SMS message delivered to a user's cellphone, in accordance with the present invention.

[0089] Referring to FIG. **7**, the User Identify Verification Method **700** depicted in FIG. **7** illustrates how a user's phone number may be verified by embodiments of the present invention. This may be necessary or desired if, for example, two-factor authentication is required for a user account.

[0090] At Step **720**, a Code may be sent to User **705** via an SMS message to a User's Phone **750** via a phone number that User **705** provided as a User Identifier **222** (see FIG. **2** described above). The

Code and SMS message may be prepared and sent using any industry standard mechanism or service known to those skilled in the art.

[0091] When the transmitted SMS code is received by User **705** on the User's Phone **750** via the user's phone number, the User **705** may enter the received code using an interface provided on a Web Page **713** accessed via Computing Device **710**. Computing Device **710** may be identical to the User's Phone **750** or it may be a separate computing device. When the User **705** enters the received SMS code into Web Page **713** within a specified period of time, Step **730** may receive the SMS code compare it to the one that was sent. If there is a match, Step **730** may then invoke Step **740** to record the fact that the identity of User **605** has been verified. Such verification may be manifested by a message transmitted to the user and/or by recording an indication of user identity verification at a corresponding security level within the Database Management System **105** described above with respect to FIG. **1** and/or DBMS **200** described above with respect to FIG. **2**

User Identity Verification Using a Trusted Name-Address Source

[0092] FIG. **8** is a combined flow chart and block diagram illustrating an exemplary embodiment of a method for verifying a user's identity to enable secure Internet communications using a trusted name-address source, in accordance with the present invention.

[0093] Referring to FIG. **8**, the User Identify Verification Method **800** depicted in FIG. **8** illustrates how trusted third-party sources may be used to verify the identity of a user such as User **805**. Such third-party sources may be used to verifiably link the real name and physical address of User **805** with electronic User Identifiers **222** (see FIG. **2**, examples of which are an email address and a phone number) that have been verified as described above. Third-Party verification may be used to establish higher security levels.

[0094] User **805** may use Computing Device **810** to create an account on the Communications Server **100** shown in FIG. **1**. During the account-creation process, User **805** may provide user name and address information at Step **820**. The user name and address information may be saved in User Verification Info **221** of DBMS **200** (see FIG. **2**) described above.

[0095] At Step **830**, the User Identity Verification Method **800** may then obtain User Name and Addresses information from a Trusted Name-Address Source **860**. This Step **830** may involve accessing more than one Trusted Name-Address Source **860**. For example, Step **830** may include reading a trusted government document such as a driver's license or state identity card. It may involve communicating with trusted data services such as Ekata (see <https://ekata.com/>). These sources may be contacted in an iterative or sequential fashion, depending on the results obtained from the next Step **840** and the specific security level being sought by User **805**.

[0096] At Step **840**, the User Identity Verification Method **800** may compare user name and address information received from User **805** with corresponding User Name and Addresses information received from a Trusted Name-Address Source **860**. If there is a substantive match, User **805** may be accepted for subsequent communications at an increased security level at Step **850**.

[0097] If there is not an adequate substantive match between user name and address information received from User **805** and corresponding User Name and Addresses information received from a Trusted Name-Address Source **860**, then Step **830** may be repeated using a different Trusted Name-Address Source **860**. Or User **805** may be denied for subsequent communications at an increased security level at Step **850**.

[0098] Because these trusted sources such as Trusted Name-Address Source **860** are neither authoritative nor authoritatively verifiable (there is no authoritative central registry for persons, email, or cell phone in the United States and the ability for non-law-enforcement personnel to verify a driver's license or passport with the issuing authority is severely restricted) mechanisms for coping with partial matches, for example, where only one of the Received User Name or Received Address matches the Trusted User Name and Trusted Address, are implemented. These coping mechanisms may consist of requiring the User **805** to select from a list of possible addresses or possible names where one member in the list contains the actual Trusted Name or Trusted Address.

The coping mechanisms may also consist of geo-location testing of the User **805** and User Device **810** against the Trusted Provider Address.

[0099] Successful completion (i.e., substantive matching) of the comparison of Received Name-Address to Trusted Name-Address at Step **840** may result in the User's Identity being Verified for higher security level at Step **850**. This may result in a message sent to the user. It may result in storing security level information in User Verification Info **221** of DBMS **200** (see FIG. 2) described above.

Encrypt

[0100] FIG. **9** is a combined flow chart and block diagram illustrating an exemplary embodiment of a method for encrypting user data, in accordance with the present invention.

[0101] FIG. **9** also illustrates the Communication Server **990** (discussed previously as Communication Server **100** in FIG. 1), User Device **910** (discussed previously as User Device **170** in FIG. 1), and the Database Management System **995** (discussed previously as Database Management System **105** in FIG. 1).

[0102] Referring to FIG. **9**, User Device **910** may correspond to User Device **170** and/or other similar User Devices disclosed and described herein. It may include User Application **920**, and Trifold **930**.

[0103] User Application **920** may comprise a software application configured to communicate with Trifold **930** to invoke the Encryption Service **940** of Trifold **930**.

[0104] Trifold **930** may correspond to Trifold **173** and/or other similar Trifolds disclosed and described herein.

[0105] Encryption Service **940** may begin at Step **941** when User Application **920** communicates with Trifold **930** to invoke Encryption Service **940**. When Encryption Service **940** is invoked, User Application **920** may provide Trifold **930** the data to encrypt and any Target User Identifier **950** for whom encryption is to be performed. The data to encrypt may be represented as a raw byte array. The Target User Identifier **950** may comprise an email addresses or the like.

[0106] At Step **943**, Encryption Service **940** may send the Target User Identifier **950** to the Communications Server **990** along with a request to return a Public Key **955** associated with that Target User Identifier **950**. When Communication Server **990** provides the Public Key **955**, Step **945** may then begin.

[0107] At Step **945**, Encryption Service **940** may encrypt the data provided by User Application **920** using any one of many industry standard cryptographic encryption functions known by those skilled in the art, using Public Key **955**. The encrypted data may be represented as an array of bytes.

[0108] At Step **947**, the encrypted data may be returned to User Application **920**.

Decrypt

[0109] FIG. **10** is a combined flow chart and block diagram illustrating an exemplary embodiment of a method for decrypting user data, in accordance with the present invention.

[0110] Referring to FIG. **10**, User Device **1010** may correspond to User Device **170** and/or other similar User Devices disclosed and described herein. It may include User Application **1020**, Trifold **1030**, and the Secure Storage **1080**.

[0111] User Application **1020** may comprise a software application configured to communicate with Trifold **1030** to invoke the Decryption Service **1040** of Trifold **1030**.

[0112] Trifold **1030** may correspond to Trifold **173** and/or other similar Trifolds disclosed and described herein.

[0113] Secure Storage **1080** may correspond to Secure Storage **540** and/or other similar Secure Storage areas disclosed and described herein.

[0114] Decryption Service **1040** may begin at Step **1041** when User Application **1020** communicates with Trifold **1030** to invoke Decryption Service **1040** by sending a Decryption Request. When Decryption Service **1040** receives the Decryption Request, User Application **1020**

may provide Trifold **1030** the data to decrypt. The data to encrypt may be represented as a raw byte array.

[0115] At Step **1043**, Decryption Service **1040** may retrieve Private Key **1082** from Secure Storage **1080**. Private Key **1082** was created by Trifold **1030** at Step **531** (Initialize) for the purpose of decrypting data that has been encrypted with a corresponding public key such as Public Key **580** discussed in the context of FIG. 5.

[0116] At Step **1045**, Decryption Service **1040** may decrypt the provided data using Private Key **1082** using any one of many industry standard cryptographic decryption functions known by those skilled in the art.

[0117] At Step **1047**, Decryption Service **1040** may return the decrypted data to User Application **1020**.

Sign

[0118] FIG. **11** is a combined flow chart and block diagram illustrating an exemplary embodiment of a method for cryptographically signing a digital document, in accordance with the present invention.

[0119] Referring to FIG. **11**, User Device **1110** may correspond to User Device **170** and/or other similar User Devices disclosed and described herein. It may include User Application **1120**, Trifold **1130**, and the Secure Storage **1180**.

[0120] User Application **1120** may comprise a software application configured to communicate with Trifold **1130** to invoke the Signing Service **1140** of Trifold **1130**.

[0121] Trifold **1130** may correspond to Trifold **173** and/or other similar Trifolds disclosed and described herein.

[0122] Secure Storage **1180** may correspond to Secure Storage **540** and/or other similar Secure Storage areas disclosed and described herein.

[0123] Signing Service **1140** may begin at Step **1141** when User Application **1120** communicates with Trifold **1130** to invoke Signing Service **1140** by sending a Signing Request. When Signing Service **1140** receives the Signing Request, User Application **1120** may provide Trifold **1130** the data or digital document to sign. The data or document to sign may be represented as a raw byte array.

[0124] At Step **1143**, Signing Service **1140** may retrieve Private Key **1182** from Secure Storage **1180**. Private Key **1182** was created by Trifold **1030** at Step **531** (Initialize) for the purpose of signing digital data, where the resulting signature can be verified with a corresponding public key such as Public Key **580** discussed in the context of FIG. 5.

[0125] At Step **1145**, Signing Service **1140** may sign the provided data using Private Key **1182** to create a digital signature using any one of many industry standard cryptographic signing functions known by those skilled in the art.

[0126] At Step **1147**, Signing Service **1140** may return the digital signature to User Application **1120**.

Verify

[0127] FIG. **12** is a combined flow chart and block diagram illustrating an exemplary embodiment of a method for verifying a digital signature, in accordance with the present invention.

[0128] FIG. **12** also illustrates the Communication Server **1290** (discussed previously as Communication Server **100** in FIG. 1), User Device **1210** (discussed previously as User Device **170** in FIG. 1), and the Database Management System **1295** (discussed previously as Database Management System **105** in FIG. 1).

[0129] Referring to FIG. **12**, User Device **1210** may correspond to User Device **170** and/or other similar User Devices disclosed and described herein. It may include User Application **1220**, and Trifold **1230**.

[0130] User Application **1220** may comprise a software application configured to communicate with Trifold **1230** to invoke the Verification Service **1240** of Trifold **1230**.

[0131] Trifold **1230** may correspond to Trifold **173** and other similar Trifolds disclosed and described herein.

[0132] Verification Service **1240** may begin at Step **1241** when User Application **1220** communicates with Trifold **1230** to invoke Verification Service **1240**. When Verification Service **1240** is invoked and receives a Verification Request, User Application **1220** may provide Trifold **1230** the data to verify and a Target User Identifier **1250** for whom verification is to be performed. The data to encrypt may be represented as a raw byte array. The Target User Identifier **1250** may comprise an email addresses or the like. The Verification Request received at Step **1241** from User Application **1220** may include a digital signature, data that was signed, and a User Identifier of the purported signer. Both the digital signature and the signed data may be represented as arrays of bytes.

[0133] At Step **1243**, Verification Service **1240** may send the Target User Identifier **1250** to the Communications Server **1290** along with a request to return a Public Key **1255** associated with that Target User Identifier **1250**. When Communication Server **1290** provides the Public Key **1255**, Step **1245** may then begin.

[0134] At Step **1245**, Public Key **1255** may be combined with the digital signature and the data that was signed. Industry standard cryptographic signature verification functions may then be used to prove that the signature either was created (verified) or was not created (not verified) using the private key associated with this Public Key **1255**.

[0135] At Step **1247**, the verification result may be returned to User Application **1220**.

Benefits and Advantages of the Embodiments

[0136] This invention brings the power of digital signature signing and verification for purposes of authentication and non-repudiation and encryption-decryption for enhanced security within reach of ordinary users. They can realize these capabilities without having to navigate the signing requests of certificate authorities, or determine how to install certificates onto a variety of different physical devices. These users are never obliged to share their private keys with other servers where they are vulnerable to a targeted attack, nor are they required to entrust confidential documents to security promised on a third party server. Their data and their security keys are always under control of software running on their personnel devices.

[0137] These capabilities are available to any application that is secured by the techniques disclosed by embodiments of the present invention and are therefore “TripleEnabled” and can take advantage of the capabilities.

[0138] Users of embodiments of the present invention also gain a much higher degree of certainty with respect to whom they are communicating on the Internet. This occurs in several dimensions. Most simply they are getting a digitally signed email from a particular address that can be easily traced to an owner. Embodiments can dramatically reduce the address spoofing that is often used in phishing attacks. On a more sophisticated level, secured communications such as are enabled by embodiments of the present invention can be tied back to a real, verified physical identity. A message received from email address user_name@domain.com will thereby not only be known to have come from that email address, but it can be easily known that the person behind the email address has been verified to be a certain person living at a specific address. Reported misuse of this credential can easily be traced and the keys used for it can be invalidated.

Computing Device

[0139] FIG. **13** is a block diagram of an exemplary embodiment of a Computing Device **1300**, in accordance with the present invention, which in certain operative embodiments can comprise, for example, the Computing Server **100**, the User Device **170**, Requester Device **180**, User Device **370**, User Device **470**, User Device **510**, Communication Server **550**, User Device **620**, User's Phone **750**, Computing Device **710**, Computing Device **810**, User Device **910**, Communications Server **990**, User Device **1010**, User Device **1110**, User Device **1210**, and Communications Server **1290**. Computing Device **1300** may comprise any of numerous components, such as for example,

one or more Network Interfaces **1310**, one or more Memories **1320**, one or more Processors **1330**, program Instructions and Logic **1340**, one or more Input/Output (“I/O”) Devices **1350**, and one or more User Interfaces **1360** that may be coupled to the I/O Device(s) **1350**, etc.

[0140] Computing Device **1300** may comprise any device known in the art that is capable of processing data and/or information, such as any general purpose and/or special purpose computer, including as a personal computer, workstation, server, minicomputer, mainframe, supercomputer, computer terminal, laptop, tablet computer (such as an iPad), wearable computer, mobile terminal, Bluetooth device, communicator, smart phone (such as an iPhone, Android device, or BlackBerry), a programmed microprocessor or microcontroller and/or peripheral integrated circuit elements, an ASIC or other integrated circuit, a hardware electronic logic circuit such as a discrete element circuit, and/or a programmable logic device such as a PLD, PLA, FPGA, or PAL, or the like, etc. In general, any device on which a finite state machine resides that is capable of implementing at least a portion of the methods, structures, API, and/or interfaces described herein may comprise Computing Device **1300**. Such a Computing Device **1300** can comprise components such as one or more Network Interfaces **1310**, one or more Processors **1330**, one or more Memories **1320** containing Instructions and Logic **1340**, one or more Input/Output (I/O) Devices **1350**, and one or more User Interfaces **1360** coupled to the I/O Devices **1350**, etc.

[0141] Memory **1320** can be any type of apparatus known in the art that is capable of storing analog or digital information, such as instructions and/or data. Examples include a non-volatile memory, volatile memory, Random Access Memory, RAM, Read Only Memory, ROM, flash memory, magnetic media, hard disk, solid state drive, floppy disk, magnetic tape, optical media, optical disk, compact disk, CD, digital versatile disk, DVD, and/or RAID array, etc. The memory device can be coupled to a processor and/or can store instructions adapted to be executed by processor, such as according to an embodiment disclosed herein.

[0142] Input/Output (I/O) Device **1350** may comprise any sensory-oriented input and/or output device known in the art, such as an audio, visual, and/or haptic device, including, for example, a monitor, display, projector, overhead display, keyboard, keypad, mouse, trackball, joystick, gamepad, wheel, touchpad, touch panel, pointing device, microphone, speaker, video camera, camera, scanner, printer, vibrator, tactile simulator, and/or tactile pad, optionally including a communications port for communication with other components in Computing Device **200**.

[0143] Instructions and Logic **1340** may comprise directions adapted to cause a machine, such as Computing Device **200**, to perform one or more particular activities, operations, or functions. The directions, which can sometimes comprise an entity called a “kernel”, “operating system”, “program”, “application”, “utility”, “subroutine”, “script”, “macro”, “file”, “project”, “module”, “library”, “class”, “object”, or “Application Programming Interface,” etc., can be embodied as machine code, source code, object code, compiled code, assembled code, interpretable code, and/or executable code, etc., in hardware, firmware, and/or software. Instructions and Logic **1340** may reside in Processor **1330** and/or Memory **1320**.

[0144] Network Interface **1310** may comprise any device, system, or subsystem capable of coupling an information device to a network. For example, Network Interface **1310** can comprise a telephone, cellular phone, cellular modem, telephone data modem, fax modem, wireless transceiver, Ethernet circuit, cable modem, digital subscriber line interface, bridge, hub, router, switch, or other similar device.

[0145] Processor **1330** may comprise a device and/or set of machine-readable instructions for performing one or more predetermined tasks. A processor can comprise any one or a combination of hardware, firmware, and/or software. A processor can utilize mechanical, pneumatic, hydraulic, electrical, magnetic, optical, informational, chemical, and/or biological principles, signals, and/or inputs to perform the task(s). In certain embodiments, a processor can act upon information by manipulating, analyzing, modifying, converting, transmitting the information for use by an executable procedure and/or an information device, and/or routing the information to an output

device. A processor can function as a central processing unit, local controller, remote controller, parallel controller, and/or distributed controller, etc. Unless stated otherwise, the processor can comprise a general-purpose device, such as a microcontroller and/or a microprocessor, such the Pentium IV series of microprocessors manufactured by the Intel Corporation of Santa Clara, California. In certain embodiments, the processor can be dedicated purpose device, such as an Application Specific Integrated Circuit (ASIC) or a Field Programmable Gate Array (FPGA) that has been designed to implement in its hardware and/or firmware at least a part of an embodiment disclosed herein.

[0146] User Interface **1360** may comprise any device and/or means for rendering information to a user and/or requesting information from the user. User Interface **1360** may include, for example, at least one of textual, graphical, audio, video, animation, and/or haptic elements. A textual element can be provided, for example, by a printer, monitor, display, projector, etc. A graphical element can be provided, for example, via a monitor, display, projector, and/or visual indication device, such as a light, flag, beacon, etc. An audio element can be provided, for example, via a speaker, microphone, and/or other sound generating and/or receiving device. A video element or animation element can be provided, for example, via a monitor, display, projector, and/or another visual device. A haptic element can be provided, for example, via a very low frequency speaker, vibrator, tactile stimulator, tactile pad, simulator, keyboard, keypad, mouse, trackball, joystick, gamepad, wheel, touchpad, touch panel, pointing device, and/or other haptic device, etc. A user interface can include one or more textual elements such as, for example, one or more letters, number, symbols, etc. A user interface can include one or more graphical elements such as, for example, an image, photograph, drawing, icon, window, title bar, panel, sheet, tab, drawer, matrix, table, form, calendar, outline view, frame, dialog box, static text, text box, list, pick list, pop-up list, pull-down list, menu, tool bar, dock, check box, radio button, hyperlink, browser, button, control, palette, preview panel, color wheel, dial, slider, scroll bar, cursor, status bar, stepper, and/or progress indicator, etc. A textual and/or graphical element can be used for selecting, programming, adjusting, changing, specifying, etc. an appearance, background color, background style, border style, border thickness, foreground color, font, font style, font size, alignment, line spacing, indent, maximum data length, validation, query, cursor type, pointer type, auto-sizing, position, and/or dimension, etc. A user interface can include one or more audio elements such as, for example, a volume control, pitch control, speed control, voice selector, and/or one or more elements for controlling audio play, speed, pause, fast forward, reverse, etc. A user interface can include one or more video elements such as, for example, elements controlling video play, speed, pause, fast forward, reverse, zoom-in, zoom-out, rotate, and/or tilt, etc. A user interface can include one or more animation elements such as, for example, elements controlling animation play, pause, fast forward, reverse, zoom-in, zoom-out, rotate, tilt, color, intensity, speed, frequency, appearance, etc. A user interface can include one or more haptic elements such as, for example, elements utilizing tactile stimulus, force, pressure, vibration, motion, displacement, temperature, etc.

VARIATIONS

[0147] The present invention can be realized in hardware, software, or a combination of hardware and software. The invention can be realized in a centralized fashion in one computer system, or in a distributed fashion where different elements are spread across several computer systems. Any kind of computer system or other apparatus adapted for carrying out the methods described herein is suitable. A typical combination of hardware and software can be a general-purpose computer system with a computer program that, when being loaded and executed, controls the computer system such that it carries out the methods described herein.

[0148] Although the present disclosure provides certain embodiments and applications, other embodiments apparent to those of ordinary skill in the art, including embodiments that do not provide all of the features and advantages set forth herein, are also within the scope of this disclosure.

[0149] The present invention, as already noted, can be embedded in a computer program product, such as a computer-readable storage medium or device which when loaded into a computer system is able to carry out the different methods described herein. A “computer program,” in the present context, means any expression, in any computing language, code or notation, of a set of instructions intended to cause a system having an information processing capability to perform a particular set of functions, either directly or indirectly, including after either or both of the following: a) conversion to another language, code or notation; or b) reproduction, including reproduction in a different material form.

[0150] The foregoing disclosure has been set forth merely to illustrate the invention and is not intended to be limiting. It will be appreciated that modifications, variations, and additional embodiments are covered by the above teachings and within the purview of the appended claims without departing from the spirit and intended scope of the invention. Other logic may also be provided as part of the exemplary embodiments but are not included here so as not to obfuscate the present invention. Since modifications of the disclosed embodiments incorporating the spirit and substance of the invention may occur to persons skilled in the art, the invention should be construed to include everything within the scope of the appended claims and equivalents thereof.

Claims

1. A system comprising: an asymmetric key manager configured to generate a key pair including a first key pair having a first private key and a paired first public key, wherein the first public key is assigned a security level to facilitate customization of secure communication, the assigned security level having a setting selected from one of public, private, secure, and enhanced; the asymmetric key manager configured to store the generated first private key in a secure storage local to a user device; and one or more service modules, operatively coupled to the asymmetric key manager and operating local to the user device, to retrieve and use the first private key stored local to the user device in support of one or more cryptographic service functions local to the user device and responsive to the assigned security level, the cryptographic functions including one or more of encryption, decryption, digital signature generation, and digital signature verification.
2. The system of claim 1, further comprising the asymmetric key manager configured to receive a transport layer security (TLS) certificate directly associated with the asymmetric key manager for the user device, wherein the TLS certificate is encrypted with a password private key, and further comprising the asymmetric key manager configured to store the TLS certificate password in the secure storage local to the user device.
3. The system of claim 2, wherein the TLS certificate is directly associated with a specific application operating local to the user device.
4. The system of claim 1, further comprising a signing service, local to the user device, operatively coupled to the asymmetric key manager, the signing service configured to apply a signature to designated digital data and further comprises: retrieval of the first private key from the secure storage local to the user device; and use the retrieved first private key to create a first digital signature for the designated digital data.
5. The system of claim 4, further comprising a digital signature verification service, local to the user device, operatively coupled to the asymmetric key manager, the verification service configured to receive a second public encryption key and selectively verify creation of the first digital signature.
6. The system of claim 5, further comprising the digital signature verification service configured to verify the assigned security level of the second public key corresponding to a result of the selectively verified first digital signature.
7. The system of claim 6, wherein assignment of the security level setting to the second public key enables security of communications to control access to an application.

- 8.** The system of claim 1, further comprising a cryptographic chain services module, operatively coupled to the asymmetric key manager, configured to store cryptographic chain data directly associated with the generated first asymmetric key pair to a remote append-only immutable record.
- 9.** A computer implemented method comprising: generating, via an asymmetric key manager, a key pair including a first key pair having a first private key and a paired first public key; assigning the first public key a security level to facilitate customization of secure communication, the assigned security level having a setting selected from one of public, private, secure, and enhanced; storing the generated first private key in a secure storage local to a user device; and retrieving and using the first private key stored local to the user device in support of one or more cryptographic service functions local to the user device and responsive to the assigned security level, the cryptographic functions including one or more of encryption, decryption, digital signature generation, and digital signature verification.
- 10.** The computer implemented method of claim 9, further comprising receiving a transport layer security (TLS) certificate directly associated with the asymmetric key manager for the user device, wherein the TLS certificate is encrypted with a password private key, and further comprising storing the TLS certificate password in the secure storage local to the user device.
- 11.** The computer implemented method of claim 10, wherein the TLS certificate is directly associated with a specific application operating local to the user device.
- 12.** The computer implemented method of claim 9, further comprising applying a signature to designated digital data, including: retrieving the first private key from the secure storage local to the user device; and using the retrieved first private key, creating a first digital signature for the designated digital data.
- 13.** The computer implemented method of claim 12, further comprising receive a second public encryption key and selectively verifying creation of the first digital signature with the second public encryption key.
- 14.** The computer implemented method of claim 13, further comprising verifying the assigned security level of the second public key corresponding to a result of the selectively verified first digital signature.
- 15.** The computer implemented method of claim 14, wherein assignment of the security level setting to the second public key enables security of communications to control access to an application.
- 16.** The computer implemented method of claim 9, further comprising storing cryptographic chain data directly associated with the generated first asymmetric key pair to a remote append-only immutable record.
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